

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – SHORT ANSWER TEST (30 Questions)

Course Code:	ITA0502	Course Title:	COMPUTER VISION
CO Assessed:	CO1 – Imitation (BL3)	Assessment:	Assessment Tool 1 – Short Answer Test
Weightage:	40% of CIA	Total Marks:	50 (10 Questions × 5 Marks)
CO4 Statement:	Apply the fundamental concepts, image formation models, and processing techniques for analyzing Computer Vision systems. (BTL 3)		
Student Name:		Reg. No.:	
Section / Batch:		Date:	

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- This test contains 6 questions, each carrying 5 marks. Total: 30 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

Section / Topic	Focus	Q Nos	Marks
Overview of Computer Vision	Definition, objectives, workflow, and importance of Computer Vision	1	5
Features of Computer Vision	Feature extraction, object recognition, and image understanding	2	5
Levels of Computer Vision and Image Processing	Low-level, mid-level, and high-level image processing operations	3	5
Digital Image Fundamentals	Pixels, image representation, grayscale/color images, and image resolution	4,10	10
Image Sensing and Acquisition	Image sensors, acquisition process, and imaging devices	5	5
Sampling and Quantization	Sampling, quantization, and digital image formation	6	5
Image Formation Models	Perspective projection, illumination, and image formation process	7	5
Applications of Image Processing and Computer Vision	Healthcare, autonomous vehicles, surveillance, agriculture, robotics, and industrial automation	8,9	10
GRAND TOTAL:		50 Marks	

Annexure A – Short Answer Test

1. How does Computer Vision enable computers to interpret and understand visual information?

- a) What are the main objectives of Computer Vision?
- b) Mention any two real-world applications of Computer Vision.

2. What are the key features of Computer Vision that make it suitable for intelligent image analysis?

- a) Define feature extraction.
- b) How does object recognition improve Computer Vision systems?

3. How are the different levels of image processing used in Computer Vision?

- a) What is low-level image processing?
- b) Differentiate between mid-level and high-level image processing.

4. How is a digital image represented and what are its fundamental components?

- a) Define a pixel.
- b) How does image resolution affect image quality?

5. How does image sensing and acquisition convert a real-world scene into a digital image?

- a) What is the role of an image sensor?
- b) Name any two image acquisition devices.

6. How do sampling and quantization contribute to the formation of a digital image?

- a) Define sampling.
- b) What is the purpose of quantization in image processing?

7. How does the image formation model describe the process of capturing images?

- a) What is perspective projection?
- b) What is the role of illumination in image formation?

8. How are image processing and Computer Vision related, and how do they differ?

- a) Define image processing.
- b) State two differences between image processing and Computer Vision.

9. How is Computer Vision applied in solving real-world problems across different domains?

- a) Explain one application in healthcare.
- b) Explain one application in autonomous vehicles.

10. How do digital image fundamentals support Computer Vision applications?

- a) Why are pixels considered the basic building blocks of digital images?
- b) How do sampling and image resolution influence image analysis?

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks	
Understanding of Concepts	50%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions		
Presentation	50%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation, lacks depth	Poor or unclear explanation		

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____